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INFRASTRUCTURE

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Infrastructural development means the development of many support facilities. Infrastructure is a major sector that propels overall development of the Indian economy. These facilities may be divided into (a)economic infrastructure- transport, communication, energy, irrigation, monetary and financial institutions and (b) social infrastructure includes education, training and research, health, housing and civic amenities.

Economic Infrastructure

Economic infrastructure is the support system which helps in facilitating production and distribution. For instance, railways, trucks, posts and telegraph offices, ports, canals, power plants, banks, insurance companies etc. are all economic infrastructure of an economy. They help in the production of goods and services.

Why Infrastructure Development is necessary

1. For a massive country such as India, improvement in infrastructure is a necessity.
2. Target of 5 trillion economy.
3. Infrastructure development will generate growth, employment and pull people out of poverty.
4. Infrastructure development will benefit Government's Ease of Doing Business.
5. Developing Renewable Energy sector will help in mitigating climate.
6. Infrastructure investments can also help improve peace and security by enabling, sustaining and enhancing societal living conditions.

TRANSPORT

For the sustained economic growth of a country, a well-connected and efficient transport system is needed. India has a good network of rail, road, coastal shipping, and air transport.

1.Roads:

Roads are part of an integrated multimodal system of transport which provides crucial links to airports, railway stations, ports and other logistical hubs and acts as a catalyst for economic growth by playing a critical role in the supply chain management. India has a road network of about 58.98 lakh kms as on 31 March, 2017 with rural roads constituting 70.65 per cent and National highways constituting 1.94 per cent India has one of the largest road networks in the world. Accounts for about 3.14 percent of GVA and 69 per cent and 90 per cent of the country wide freight and passenger traffic respectively. Major outcomes in Road sector during the period 2014-15 to 2018-19 were construction of Eastern Peripheral Expressway, Delhi-Meerut expressway and Dhola-Sadiya Bridge.

2.Railways:

The history of rail transport in India dates back to 1832 at madras. India has a broad network of railroad lines, the largest in Asia and the fourth largest in the world. Being a cost-effective, long distant transport mode, provide safe, secure and comfortable journey to passengers. Indian Railways Provide Wi-Fi Facility First in India is Bangalore Railway Station. Indian railways has initiated a major electrification program for electrifying 100 per cent of its Broad Gauge network also reduce the nation's dependence on imported diesel oil. Indian Railways has 35,488 Route Kilometer (RKM) of network commissioned on electric traction. Swachh Rail, Swachh Bharat', mission focuses on cleanliness.

3.Water Ways:

Shipping plays a pivotal role in India's Trade. Ports handle around 90 percent of EXIM Cargo by volume and 70 percent by value. The ports including Calcutta, Mumbai, Chennai, Vishakhapatnam and Goa are visited by cargo carriers and passenger liners from all parts of the world. The National Harbour board was set up in1950 to advise the Central and State Governments on the management and development of ports, particularly minor ports. India's first inland waterway multimodal terminal (MMT) at Varanasi was inaugurated on 12 November 2018. The cargo traffic on National Waterways was 55 million tonnes in 2017-18 and has increased by 31 per cent in 2018-19.

4.Airways:

Comprehensive network of air routes connects the major cities and towns of the country. The domestic air services are being looked after by Indian Airlines and private airlines. The international airport service is looked after by Air India. Air India and Indian Airlines were merged on August 27, 2007 to from National Aviation Company of India Ltd. (NACIL). India's scheduled domestic air transportation for passengers and goods has grown by 14 per cent and 12 per cent respectively in 2018-19. Domestic air cargo growth of 12.1 per cent in 2018-19.

ENERGY

Energy is one of the necessary components of our life. Nowadays, without electricity, we cannot survive in this world of technology. The energy sources are classified under two heads based on the availability of the raw materials used, while generating energy.

1. Non-renewable energy sources
2. Renewable energy sources

1. Non-renewable energy sources

The sources of energy which cannot be renewed or re-used are called non renewable energy sources. Basically, these are the energy sources which will get exhausted over a period of time. The examples of this kind of resources are coal, oil, gas etc.

1. Coal

Coal is an inflammable organic substance composed mainly of hydrocarbons. Coal is available in the form of sedimentary rocks. It is used in the generation of thermal power. It has close association with the industrial development of any country. Since it is a valuable one, it is called as "Black Gold". Based on carbon content, it is classified in to the following types.

Anthracite: contains 80 to 90% carbon

Bituminous: contains 60 to 80% carbon

Lignite: contains 40 to 60% carbon

Peat: contains less than 40% carbon. Coal deposits are mainly confined to eastern and south-central parts of the country. The states of Jharkhand, Odisha, Chhattisgarh, West Bengal, Madhya Pradesh, Telangana and Maharashtra account for 98.26% of the total coal reserves in the country. The State of

Jharkhand had the maximum share (26.06%) in the overall reserves of coal in the country followed by the State of Odisha (24.86%).

2. Lignite

Indian lignite (brown coal) deposits occur in the southern and western parts of Peninsular

India particularly in Tamilnadu, Puducherry and Kerala. The estimated total reserves of lignite as on 2018 was 45.66 billion tonnes.

3. Petroleum (or) Crude oil

Geographical distribution of Crude oil indicates that the maximum reserves are in the Western Offshore (40%) followed by Assam (27%), whereas the maximum reserves of Natural Gas are in the Eastern Offshore (38.13%) followed by Western offshore (23.33%). The estimated reserves of crude oil in India as on 2018 stood at 594.49 million tonnes (MT) against 604.10 million tonnes on 2017. There was decrease of 1.59% in the estimated reserve of crude oil for the country as a whole during 2017-18 except for Arunachal Pradesh which has shown an increase of 14.3%. The Ministry of Petroleum and Natural Gas (MOP&NG) is a ministry of the Government of India.

4. Natural Gas

It is formed when layers of decomposed plants and animals are exposed to intense heat and pressure over thousands of years. It is used as a source of energy for heating, cooking and electricity generation. It is also used as fuel for vehicles and as a chemical feedstock in the manufacture of plastics and other commercially important organic chemicals. The estimated reserves of Natural Gas increased by 3.87% over the last year. The maximum contribution to this increase has been from Arunachal Pradesh, and Rajasthan followed by Andhra Pradesh and Tamil Nadu.

2. Renewable energy sources

These are the kind of energy source which can be renewed or reused again and again. These kinds of materials do not exhaust or literally speaking these are available in

abundant or infinite quantity. Example of this this kind include Solar energy, Wind energy, Tidal energy, Geothermal energy, Biomass energy.

Power Sector

Power sector in India has witnessed a paradigm shift over the years due to the constant efforts of Government to foster investment in the sector. India improved its ranking in the Energy Transition Index published by World Economic Forum 76th position. The installed capacity has increased from3,44,002 MW in 2018 to 3,56,100.19 MW in 2019. The capacity of thermal power is 64 per cent followed by renewable energy.

Fuel	**MW**	**% of Total**
Total Thermal	2,27,644	63.2%
Coal	1,95,810	54.3%
Lignite	6,260	1.7%
Gas	24,937	6.9%
Diesel	638	0.2%
Hydro (Renewable)	45,399	12.6%
Nuclear	6,780	1.9%
RES* (MNRE)	80,633	22.0%
Total	**360,456**	

COMMUNICATION

Communication is a process that involves exchange of information, thoughts and ideas. Technology does wonders in communication fields.

Personal Communication

The exchange of information between the individuals is called personal communication. It includes post and telegraph services, telephone, mobile phone, short message services, fax, internet, e-mail etc. Personal Communication system enables the user to establish direct contact.

1. Postal

The Indian postal network is the largest in the world with 1,55,000 post offices. Of these more than 1,39,000 post offices are located in rural areas. The postal service was opened to the public in the country in 1837. The first Indian postal stamp was issued in 1852 in Karachi. Collecting and delivering mail is the primary function of the department of posts. It introduced the Quick Mail Service in 1975 and today it covers the entire country. The Quick Mail Service functions on the basis of the system of PIN (Postal Index Number) code which was introduced in 1972. The premium products include the Money order, e-money order, Speed Post, Express Parcel Post, Business Post, Media Post, Satellite Post, Retail Post, Greeting Post, Data Post, Speed Net and Speed Passport Services.

2.Telecommunication

India has one of the largest telecommunication networks in Asia. Apart from the urban areas more than two-thirds of the villages in India have already been covered with Subscriber Trunk Dialing (STD) telephone facility, while International communication can be made through ISD (International Subscriber Dialing). Telephone is a form of oral communication. It is considered very essential for the growth of commerce. It is the most preferred form as it provides instant communication. Mobile phone, fax and internet are the other personal communication used in the country. Telecommunication has been recognized world-over as a powerful tool of development and poverty reduction through empowerment of masses.

3.Mass Communication Systems

Mass Communication enables millions of people to get the information at the same time. It is a great way to provide education as well as entertainment. It helps in creating awareness among the people regarding various national policies and programmes. The

Mass Communication Systems can provide the information to people in two methods. They are Print Media and Electronic Media.

4.Electronic Media:

Radio broadcasting in India was started in 1923 by the Radio club of Bombay. Since then it gained immense popularity and changed the social and cultural life of people. It was named as All India Radio (AIR) in 1936 and again it was renamed as Akashwani in 1957. It broadcasts a variety of programs related to information, education and entertainment. Special news bulletins are also broadcasted on special occasions like session of parliament and state legislatures.

5.Television

Broadcasting has emerged as the most effective audio-visual medium for disseminating information and educating the masses. Television network in India is known as Doordarshan (DD) which started Common National Program (CNP) services and it is extended to the backward and remote rural areas.

6.Internet:

It is the global system of interconnected computer networks that use the Internet protocol suite to link devices worldwide.

7.Social media:

It is an interactive computer-mediated technology that facilitate the creation and sharing of information, ideas, career interests and other forms of expression via virtual communities and networks. With over 460 million internet users, India is the second largest online market, ranked only behind China.

8.Print Media:

Newspapers are the most common but powerful means of communication come under print media. India has many newspapers which carry information on local, national and international events to the people.

9.Satellite Communication

The use of Satellite in getting a continuous and synoptic view of larger area has made this communication system very vital for the country. Satellite images are used for

weather forecasting, monitoring of natural calamities, surveillance of border areas etc. The communication through satellites emerged as a new era in communication in our country after the establishment of Indian Space Research Organization (ISRO) in 1969. Satellite system in India can be grouped into two-the Indian National Satellite System (INSAT) and the Indian Remote Sensing Satellite System (IRS). The INSAT, established in 1983, is a multipurpose system for telecommunication, meteorological observation and for various other programs. The INSAT series are used for relaying signals to television, telephone, radio, mobile phone. It is also useful in weather detection, internet and military applications. The INSAT series, GSAT series, KALPANA-1, HAMSAT, EDUSAT are the major communication satellite used for communication purpose. GSAT–7A is the recent launch (December 19, 2018) for communication programs. INSAT-1B launched on 30th August 1983 is the first communication satellite in INSAT series.

SOCIAL INFRASTRUCTURE

Priority to social infrastructure is stated as essentials to inclusive and sustainable growth. Higher public investment in the social sector, including education and health, is critical for India. India has made significant progress in quantitative indicators such as enrolment levels and physical infrastructure like construction of school buildings, drinking water facilities, toilet, etc. The quality of education also needs to be monitored and assessed. India has been successful in achieving gender parity in the school sector and in higher education it is moving towards a better gender parity. Growing expenditure on health is burdening the public in general. Public investment in social infrastructure has a critical role in providing access to social services for the people, especially the marginal and vulnerable sections of the society. The expenditure on social services by the Centre and States as a proportion of GDP 7.3 per cent in 2018-19

Education

1. Education in India

Imparting education on an organized basis dates back to the days of 'Gurukul' in India. Since then the Indian education system has flourished and developed with the growing needs of the economy. The Ministry & Human Resource Development (MHRD) in India formulates education policy in India and also undertakes education programs. General education of youth has improved to 65.4 per cent for urban females and 65.8

per cent for urban males providing primary education has yielded results across social categories and gender in Gross Enrolment Rate (GER).

2. Education system in India

Education in India until 1976 was the responsibility of the State governments. It was then brought under concurrent list (both Centre and State). The Centre is represented by the Ministry of Human Resource Development decides the India's education budget. The education system in India consists of primarily six levels:

1. Nursery Class
2. Primary Class
3. Secondary Level
4. Higher Secondary Level
5. Graduation
6. Post-Graduation

3. Education Institutions in India:

Education in India follows the 10+2 pattern. For higher education, there are various State run as well as private institutions and universities providing a variety of courses and subjects. The accreditation of the universities is decided. Under the University Grant Commission Act. The Education Department consists of various schools, colleges and universities imparting education on fair means for all sections of the society. The budget share of the education sector is around 3% of GDP, of this largest proportion goes for school education. However, per pupil expenditure is the lowest for school students.

Health

1. Health in India

Health in India is a state government responsibility. The Central Council of Health and Welfare formulates the various health care projects and health department reform policies. The administration of health industry in India as well as the technical needs of the health sector are the responsibility of the Ministry of Health and Welfare. A healthy productive population is an enabler for sustainable development and hence it is critical that the Government remains committed to improve public health delivery, reduce health inequities and ensure affordable health care for all.

Form of Health care in India

1. Ayurvedic medicine practice,
2. Unani or galenic herbal care,
3. Homeopathy,
4. Allopathy,
5. Yoga.

Each different healthcare form has its own treatment system and practice patterns. The medical practicing in India needs a proper licensing from the Ministry of Health. All medical systems are now under one ministry viz AYUSH.

2. Health Care Services in India:

The health care services in India are mainly the responsibility of the Ministry of Health. State wise, health status is better in Kerala as compared to other States.

SCIENCE & TECHNOLOGY AND R&D

1. Strengthening & Modernization of Pest Management Approach in India (SMPMA)

Objectives

1. Maximize crop production with minimum input costs.
2. Minimize environmental pollution in soil, water and air due to pesticides.
3. Minimize occupational health hazards due to chemical pesticides.
4. It is a central sector scheme launched with following component so Integrated Pest Management (IPM) – It is environment friendly broad ecological approach for managing pest problems.

Salient Features

1. **Locust Control and Research –** under this locust warning organizations have been established to monitor, forewarn and control locust in Scheduled Desert Area (of Rajasthan, Gujarat and Haryana) and to conduct research on locust and grasshoppers.
2. **Implementation of Insecticides Act, 1968 –** It regulates the import, manufacture, sale, transport, distribution and use of insecticides with a view to prevent risk to human beings, animals and environment.

3. **Implementing Agency** - 35 Central Integrated Pest Management Centres (CIPMCs)

2. Mera Gaon-Mera Gaurav

Objective

To promote direct interface of scientists with the farmers.

Salient features

1. This scheme involves scientists of the Indian Council of Agriculture Research (ICAR) and state agricultural universities.
2. Groups of four multidisciplinary scientists each will be constituted at these institutes and universities. Each group will "adopt" five villages within a radius of maximum 100 km.
3. Under this scheme, scientists will select villages as per their convenience and will remain in touch with the selected villages and provide information to the farmers on technical and other related aspects in a time frame.

3. DARPAN (Digital Advancement of Rural Post Office for A New India) Project

1. It aims to improve the quality of service and achieve "financial inclusion" of un-banked rural population.
2. As part of IT modernization project, it intends to provide a low power technology solution to each Branch Postmaster (BPM) which will enable 1.29 Rural Lakhs Branch Post Offices (BOs).
3. Recently, DARPAN-PLI application was launched for seamless collection of premium for Postal Life Insurance (PLI) and Rural Postal Life Insurance (RPLI) policies.

4. Sampoorna Bima Gram Yojana

1. It aims to provide affordable life insurance services to people living in rural areas of the country through the postal network.
2. It will identify at least one village (having a minimum of 100 households) in each of the revenue districts of the country and cover all households with a minimum of one RPLI (Rural Postal Life Insurance) policy.

3. All villages under Sansad Adarsh Gram Yojana would also be covered under the scheme to turn them into Sampoorna Beema Gram.

5. North East Road Sector Development Scheme

Objective is to take up rehabilitation/construction/upgradation of neglected inter-state roads (including bridges on the roads) in the North Eastern Region.

Other criteria of roads to be constructed under the scheme are;

1. Roads in socio-politically neglected pockets of NER.
2. Roads required for security or strategic viewpoint, not covered in any other programmes.
3. Roads necessary from the viewpoint of market access for agriculture produce and roads of economic importance on gap filling approach.
4. The Scheme will be administered and monitored in the Ministry of DoNER through an Empowered Inter-Ministerial Committee under chairmanship of Secretary, DoNER.

6. Digital India

Objective

To transform India into a digitally empowered society and knowledge economy.

Salient features

1. The Digital India programme is centred on three key vision areas: Digital Infrastructure as a Utility to Every Citizen, Governance & Services on Demand, Digital Empowerment of Citizens
2. It aims to provide the much-needed thrust to the nine pillars of growth areas, namely:
    a) Broadband Highways,
    b) Universal Access to Mobile Connectivity,
    c) Public Internet Access Programme,
    d) e-Governance:
    e) Reforming Government through Technology,
    f) e-Kranti: NeGP 2.0,
    g) Information for All,
    h) Electronics Manufacturing,
    i) IT for Jobs, Early Harvest Programmes.

3. Public Private Partnerships would be preferred wherever feasible to implement e- Governance projects.
4. For effective management of the Digital India programme, the programme management structure would consist of a Monitoring Committee on Digital India headed by the Prime Minister, a Digital India Advisory Group chaired by the Minister of Communications and IT and an Apex Committee chaired by the Cabinet Secretary.
5. The positions of Chief Information Officers (CIO) would be created in at least 10 key Ministries so that various e-Governance projects could be designed, developed and implemented faster.
6. In order to take the benefits of Digital India to every corner of country, a vast network of more than 2.5 lakh Common Services Centers have been created. This has developed digital entrepreneurs among poor, marginalized, Dalits and women of India.

7. Bharat Interface For Money (BHIM)

Objectives

To enable fast, secure, reliable cashless payments through your mobile phone.

Salient features

1. It is an app that makes payment transactions simple, easy and quick using Unified Payments Interface (UPI). It enables direct bank to bank payments instantly and collect money using Mobile number, Bank a/c and IFSC code, Aadhaar number or Virtual Payment Address (VPA).
2. It is developed by the National Payment Corporation of India (NPCI), a not for profit company for providing retail payment systems in India under guidance from RBI.
3. It is interoperable with other Unified Payment Interface (UPI) applications, and bank accounts.
4. BHIM-Aadhaar platform is the merchant interface of the BHIM App that has been launched for making digital payments using the Aadhaar platform. Any citizen without access to smart phones, internet, debit or credit cards will be able to transact digitally through the BHIM Aadhaar platform.

Question:

1. Explain transportation systems of India briefly.